

Stage 1: Eng-Teacher Pretraining

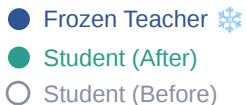
English SQuAD 2.0

XLM-R Encoder
(M_S , snowflake)

Frozen Teacher snowflake
(M_S , fixed reference)

Provides Stable Source
Representations:

- Src Geometry (Macro OT)
- Span Anchors (Micro Barycentric)
- Src Margin (Boundary Constraint)

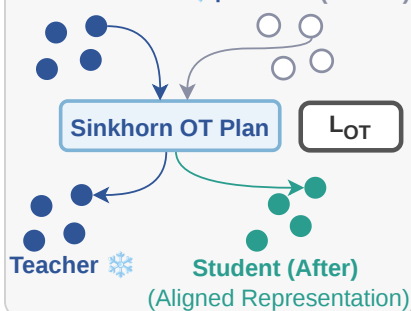


Stage 2: Coordinated Alignment Framework (Student Adaption)

1. Spatial Axis (Spatial Coordination)

1. Macro Alignment (OT Geometry)

Teacher Cluster snowflake | Student (Before)



2. Micro Alignment (y-Bary Projection)

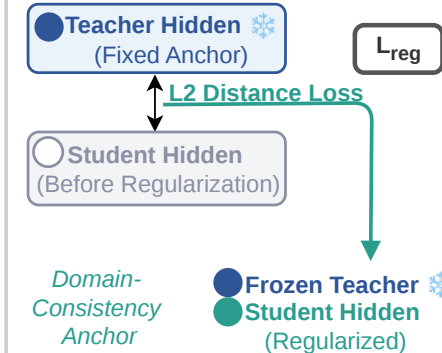
Teacher Span snowflake :
[Paris] $\xrightarrow{\text{y-Bary Projection}}$ Pseudo Span [Paris]

Student (Before):
The capital of France [is Paris in]

KL Divergence Loss L_{span}

Student (After):
The capital of France is [Paris]

2. Domain Axis (Domain Coordination)



3. Output Axis (Boundary Regularization)

Teacher Margin (Gold-Hard Neg) snowflake
Top1 [] | Top2 [] (Detached)

Student Margin (Before)
Top1 [] | Top2 [] (Small Gap)

Margin Constraint

Student Margin (After)
Top1 [] | Top2 [] (Higher Confidence Gap)

L_{margin}

